

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

1. Evaluate each of the following integrals using IBP.

a) $\int (2x+1)\cosh(x) dx$

b) $\int \frac{x}{\cos^2(x)} dx$

c) $\int_1^e \frac{\ln(x)}{\sqrt{x}} dx$

2. On which of the following would be easy to use Tabular Integration by Parts?

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| (i) $\int x^5 \sin(\pi x) dx$ | (ii) $\int x \ln(x) dx$ | (iii) $\int \tan(x) \sin(x) dx$ |
| (iv) $\int (2^x + 1) \cos(x) dx$ | (v) $\int x^2 e^x dx$ | (vi) $\int (3x + 2) \tan^{-1}(x) dx$ |

3. Evaluate $\int x^5 \sin(\pi x) dx$.

4. $\int x^3 e^{x^2} dx$ (*Hint: Start with a substitution to make things easier*)

5. Evaluate the following integral $\int e^x [\cos(x) + \sin(x)] dx$.

6. When performing integration by parts we define $dv = g(x)dx$ and then compute $v = G(x)$. It is a frequent question to ask why we don't write $v = G(x) + C$. Show that in general $uv - \int v du = u(v + C_1) - \int (v + C_1) du$ thereby justifying the omission of the constant of integration when calculating v in integration by parts)

7. Use calculus I integration techniques (differentiation rules, algebra, & u-substitutions) to answer each of the following questions.

a) Evaluate the integral $\int_0^1 (1-x^2)^9 x^9 dx$.

b) Suppose that $\int_{-\pi/2}^{\pi/2} \left[e^{\sin(x)} \cos(x) \right] dx = a - \frac{1}{a}$ where $a > 0$. Determine the value of a .

c) Evaluate the definite integral $\int \frac{dx}{1 + e^{-x}}$.

d) Let $a > 1$ be a constant satisfying the equation below. What is the value of $a^2 + a + 1$?

$$\int_1^a \frac{x-1}{x+\sqrt{x}} dx = 4$$